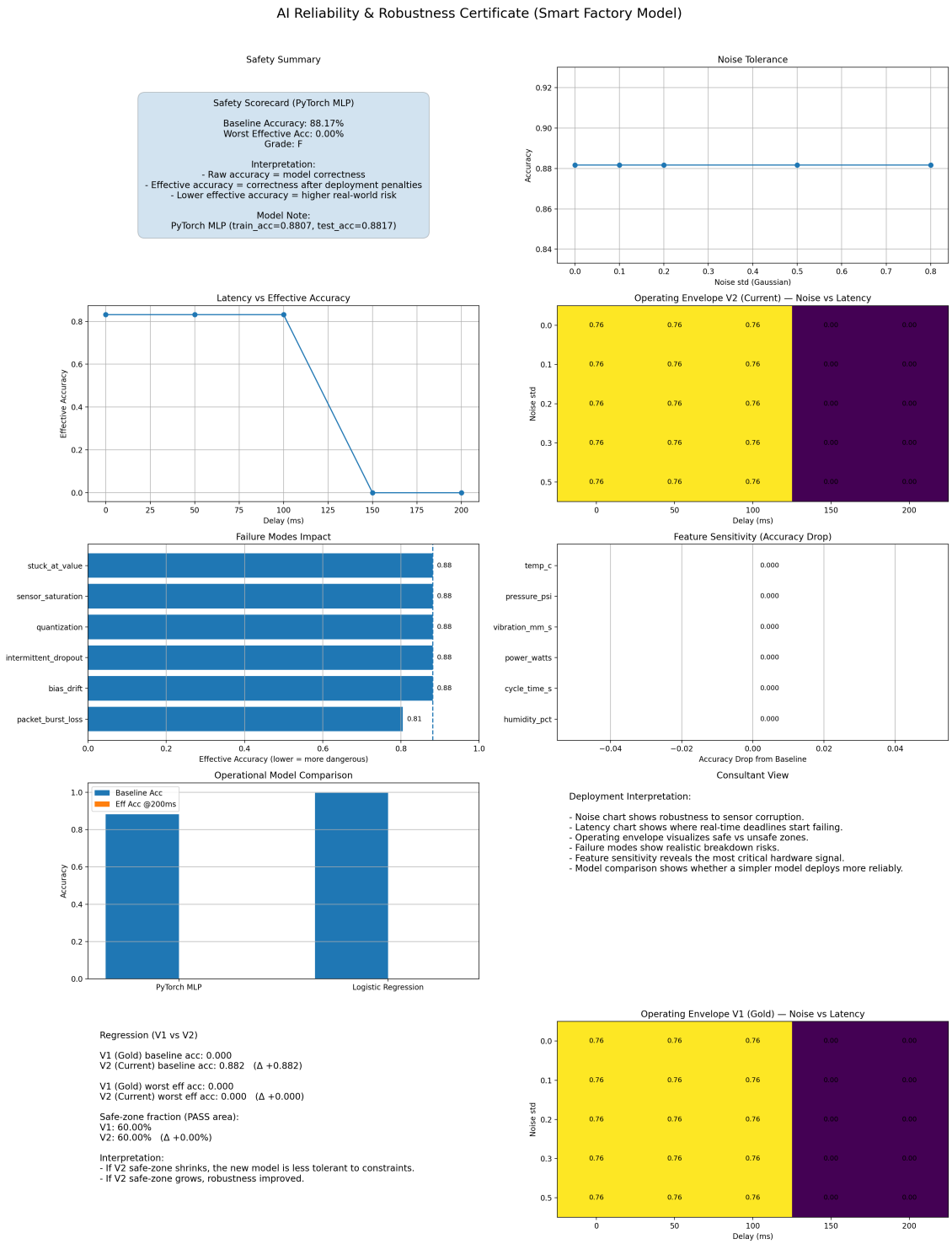
RobustAI Engine — Executive Reliability Report

Generated: 2026-03-11 14:55:08

1) Reliability Certificate Dashboard



2) Week Report

Week 06 – Smart Factory Model Reliability Report (Startup Demo)

Generated: 2026-03-11T14:55:06
Summary

Model: PyTorch MLP (trainacc=0.8807, testacc=0.8817)

Safety Grade: F (worst effective accuracy=0.0000)
Positives
• Profile: industrialwifi (threshold=0.75, timeoutms=120, droprate=0.05)
• PyTorch MLP baseline accuracy: 0.8817
• Logistic Regression baseline accuracy: 0.9967
• Includes explicit combined-constraints tests (noise + latency + staleness).
• Produced stress tests, operating envelope, regression artifacts, and prescriptive mitigations.
Issues / Concerns
• Worst effective accuracy observed (global): 0.0000 (Grade F)
• Late/missing predictions are treated as failures.
• Some latency conditions can collapse effective accuracy.
Operating Envelope

PASS/FAIL grid saved in results/week06operatingenvelope.csv.
Regression Check (V1 vs V2)

Gold profile: industrialwifi | Current profile: industrialwifi
Baseline accuracy change: +0.000
Worst effective accuracy change: +0.000
Safe operating zone change: +0.000 (fraction of envelope cells passing)
No major envelope change detected.
Required Mitigations (Fix-it Engine)
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• Option A (Software): Add a 50ms prediction buffer (queue/pipeline) to tolerate real-time jitter.
• Reason: effective accuracy fell below threshold at ~150ms (eff=0.00%, threshold=75%).
• Option B (Hardware): Improve tempc sensor stability (reduce noise by ~0.1σ).
• Reason: tempc had the largest accuracy drop during single-feature noise injection (Δ=0.000).

(Profile used: industrialwifi)
Artifacts
• Results CSV: results/week06productdemoresults.csv
• Envelope CSV (MLP): results/week06operatingenvelope.csv
• Dashboard: results/masterdashboard.png
• Mitigations: results/mitigations.md
• Regression JSON: results/regressionreport.json
• Regression Summary: results/regressionsummary.md

3) Regression Check (V1 vs V2)

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